

# Fine-mapping under a sparse genetic architecture

## Iteration 004 benchmark — complete analysis of the sparse arm

fmbenchmark

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### Abstract

This report analyses the *sparse* arm of the Iteration 004 fine-mapping benchmark: 9 design rows, 11,250 method fits per method, 18 methods, all fitted with in-sample LD. It covers overall accuracy, the conditions under which each method performs well, false-discovery and calibration behaviour, credible-set economics, and the three “senses” of variable importance (variance-explaining, decision-relevant, diagnostic). Every quantity derived from posterior inclusion probabilities is defined explicitly in §2 before it is used, including the estimator’s tie handling, its aggregation level, and the noise correction applied to the variance decomposition. The headline results are: (i) the number of causal variants  $S$  dominates the variance budget everywhere; (ii) the *presence* of annotations matters and their *strength* barely does; (iii) the Functional BEATRICE ablation is monotone and significant, with region sharing contributing more than annotations alone under continuous annotations; and (iv) the BEATRICE family buys ranking accuracy at a measurable cost in calibration.

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## 1 What was simulated, and what a “cell” is

The sparse arm holds the genetic architecture fixed at a *sparse* model: each region contains exactly  $S$  causal variants and no infinitesimal background. Nine design rows cross the annotation condition with the enrichment level:

| rows | annotation type | enrichment folds    |
|------|-----------------|---------------------|
| 1    | none            | —                   |
| 2–5  | binary          | 2.7, 5.4, 8.1, 10.8 |
| 6–9  | continuous      | 2.7, 5.4, 8.1, 10.8 |

Within every row the same sweep is run:  $S \in \{1, 2, 3, 5, 10\}$ ,  $\phi \in \{0.05, 0.1, 0.2, 0.4, 0.6\}$  (region heritability), 10 iterations, and 10 regions whose sizes are  $\{500, 500, 750, 750, 1000, 1000, 1500, 1500, 2000, 2000\}$  variants, with  $n = 5000$  individuals. That is  $5 \times 5 \times 10 = 250$  scenarios per row and  $250 \times 10 = 2,500$  fits per method per row.

**The unit of analysis.** A cell is a (row,  $S$ ,  $\phi$ , region size) combination. Each cell aggregates 10 iterations  $\times$  2 regions of that size = 20 fits. There are  $9 \times 5 \times 5 \times 5 = 1,125$  cells per method. Cell means are the input to every ANOVA in §§8–10; individual fits are the input to the calibration counts in §6. The distinction matters and is stated wherever a number is quoted.

**Two regions per size class.** Region size is deliberately duplicated. The two regions of a given size share a size class but are *different draws* from the reference panel, which is what makes the between-region variance component  $\sigma_u^2$  identifiable (§2.5). Without it the split-plot correction could not be computed at all.

**Stratification is not optional.** Results are never pooled across annotation type. A method that cannot use annotations is not comparable to one that can, and the “none” arm additionally serves as a negative control: any method whose annotation machinery is inert there must reduce *exactly* to its annotation-blind counterpart. Two such identities are checked mechanically (§11).

## 2 From posterior inclusion probabilities to numbers

Every method returns, for each region, a vector  $\text{PIP}_1, \dots, \text{PIP}_p \in [0, 1]$  — one posterior inclusion probability per variant — together with the truth  $y_j \in \{0, 1\}$  known from the simulation. All metrics below are functions of that pair. This section defines each one and justifies the estimator, because several of the obvious implementations are wrong in ways that silently change conclusions.

## 2.1 Average precision (the ranking metric)

Average precision summarises the whole precision–recall curve without committing to a threshold, which matters here because methods’ PIP scales differ by orders of magnitude: any fixed-threshold comparison partly measures calibration rather than ranking.

Sort the variants by PIP descending. Let  $k$  index the distinct PIP values (so that all variants sharing a value form one block),  $\text{tp}_k$  the cumulative number of causals at the end of block  $k$ ,  $n_k$  the cumulative number of variants,  $\text{prec}_k = \text{tp}_k/n_k$  and  $\text{rec}_k = \text{tp}_k/S$ . Then

$$\text{AP} = \sum_k (\text{rec}_k - \text{rec}_{k-1}) \text{prec}_k, \quad \text{rec}_0 = 0. \quad (1)$$

Three implementation points are load-bearing.

**It is a step integral, not a trapezoid.** Equation (1) evaluates precision at the operating point reached, matching `sklearn.metrics.average_precision_score`. A trapezoidal rule interpolates between operating points that the classifier never actually occupies and is optimistically biased.

**Ties are resolved by block, not by index.** Several methods emit many exactly-tied PIPs. Implementing AP as “sort, then read off precision at the causal’s index” lets the sort’s tie-break — here genomic position — decide the score. Because causal variants are drawn uniformly at random this would not introduce a constant bias but would inject pure noise into precisely the comparisons the analysis exists to make. Collapsing tied variants into one block gives every member the same operating point, which is the only tie-handling that is invariant to input order.

**AP is computed per fit and then averaged, never on a merged ranking.** Pooling all variants from all regions into one ranking and computing a single AP is a different and incorrect quantity: it rewards a method for getting the *relative calibration across regions* right, which is not what region-level fine-mapping is asked to do. The two disagree materially. For two fits with  $\text{AP} = 1$  and  $\text{AP} = 1/3$ , the correct answer is the mean  $2/3 \approx 0.667$ , whereas the merged ranking of the same data gives 0.700. Both values are checked against hand computation in the test suite.

A fit with no causal variants returns **NA** rather than 0, and NAs are excluded from means with the count reported alongside, so a method is never credited or penalised for a fit that posed no question.

## 2.2 Thresholded discovery: FDR and its honest companion

For a threshold  $t$  let  $\mathcal{S}_t = \{j : \text{PIP}_j \geq t\}$ ,  $n_t = |\mathcal{S}_t|$  and  $\text{tp}_t = \sum_{j \in \mathcal{S}_t} y_j$ . Then

$$\text{FDR}_t = \frac{n_t - \text{tp}_t}{n_t}, \quad \text{reported jointly with } n_t. \quad (2)$$

Reporting FDR alone is not meaningful: a method that never crosses  $t$  has  $\text{FDR}_t = 0$  by convention and  $n_t = 0$ , and would otherwise win the comparison by abstaining. The pair  $(\text{FDR}_t, n_t)$  is therefore always presented together, and  $n_t$  appears in the credible-set analysis (§2.4) as the follow-up burden.

We also compute the **honesty ratio**. A Bayesian method that emits PIP values is making a falsifiable claim about its own error rate: the expected number of false positives among  $\mathcal{S}_t$  is  $\widehat{\text{FP}}_t = \sum_{j \in \mathcal{S}_t} (1 - \text{PIP}_j)$ . Comparing that to the realised count gives

$$\rho_t = \frac{\text{FP}_t}{\widehat{\text{FP}}_t}, \quad (3)$$

so  $\rho_t = 1$  means the method’s own uncertainty statement was accurate,  $\rho_t > 1$  means it was overconfident by that factor. This is stricter than FDR: it holds the method to *its own* claim rather than to an external threshold.

### 2.3 Calibration

Two complementary summaries are used, because each is separately gameable.

**Total mass ratio.**  $\sum_j \text{PIP}_j / S$ . A coherent posterior over which variants are causal should place total mass equal to the expected number of causals. A ratio of 2 means the method claims twice as many causal variants as exist.

**Banded reliability.** Variants are binned by PIP into  $[0, 0.1)$ ,  $[0.1, 0.5)$ ,  $[0.5, 0.9)$ ,  $[0.9, 1]$ . Within a bin we accumulate the count  $n_b$ , the number truly causal  $c_b$ , and  $\sum \text{PIP}$ . Calibration compares the realised rate  $c_b/n_b$  to the claimed mean  $\sum \text{PIP}/n_b$ ; this is the reliability diagram in Figure 7. We report  $\text{ECE}_{hi}$ , the expected calibration error restricted to bins with  $\text{PIP} \geq 0.1$ , because the  $[0, 0.1)$  bin contains the overwhelming majority of variants and an unrestricted ECE is dominated by the trivially-correct statement that almost nothing is causal.

**Why both, and why RES.** Consider a predictor emitting  $\text{PIP}_j = S/p$  for every variant. Its mass ratio is exactly 1 and its calibration error is exactly 0 — it is perfectly calibrated and completely useless. The Murphy decomposition of the Brier score separates this out:

$$\text{BS} = \underbrace{\text{REL}}_{\text{miscalibration}} - \underbrace{\text{RES}}_{\text{resolution}} + \underbrace{\text{UNC}}_{\text{base rate}}, \quad (4)$$

where RES measures how far the method’s conditional rates depart from the base rate. The uniform predictor has  $\text{REL} = 0$  and  $\text{RES} = 0$ . Resolution is therefore the component that detects calibration achieved by saying nothing, and it is reported in Table 3 and Figure 9 alongside REL.

### 2.4 Credible sets

For a target mass  $\alpha$ , variants are added in decreasing PIP order until the cumulative mass first reaches  $\alpha$ ; the resulting set size is  $|CS_\alpha|$  and coverage is the indicator that the set contains a causal variant, averaged over fits. When a posterior’s *total* mass never reaches  $\alpha$  the set is the whole region and a flag `set_reached` records it — this is a result, not a missing value, and suppressing it would let a maximally diffuse posterior masquerade as a confident small set. Size and coverage are always read together (Figure 10): a set can be made to cover by being large, and small by abandoning coverage.

### 2.5 The replication structure and the noise floor

The design is a split plot. Two variance components matter:  $\sigma_\epsilon^2$ , the between-iteration variance at fixed region, and  $\sigma_u^2$ , the between-region-draw variance. A factor whose levels are compared *within* the same region draws (namely  $S$  and  $\phi$ , which are re-simulated per scenario) sees only  $\sigma_\epsilon^2$ . A factor whose levels necessarily use *different* region draws — region size, and any term containing it — additionally carries  $\sigma_u^2$ . Those are “whole-plot” terms.

For a term  $T$  with  $\text{df}_T$  degrees of freedom, the expected sum of squares under the null is  $\text{df}_T \lambda_T$  with

$$\lambda_T = \frac{\sigma_\epsilon^2}{20} + \mathbf{1}\{T \text{ whole-plot}\} \cdot \frac{m \sigma_u^2}{2}, \quad m = 25, \quad (5)$$

the 20 being the fits per cell and the 2 the regions per size class. The noise-corrected share is

$$\omega_T^2 = \frac{\text{SS}_T - \text{df}_T \lambda_T}{\text{SS}_{\text{tot}}}. \quad (6)$$

**Why the correction is necessary.** Without it, a factor that explains nothing still receives a share proportional to its degrees of freedom, and whole-plot terms receive a large one. In simulation, an uncorrected null factor recorded a 14.6% share where the truth was zero; corrected,

it averaged  $-0.6\%$  over replicates. Negative values are reported as computed and never clamped: a systematically negative share is a diagnostic that  $\hat{\lambda}$  is too large, and hiding it would hide the diagnostic.

**Estimating  $\sigma_u^2$  requires a bias correction.** The naive pooled estimator over  $J$  region draws in  $K$  groups is biased low by the factor  $c = K(J - 1)/(JK - 1) = 0.758$  at  $J = 4$  — because the pooled mean absorbs part of the between-draw signal. Dividing by  $c$  restores unbiasedness; verified over 120 replicates, recovering 0.0438 against a true 0.0400 where the naive estimator gave 0.0332.

## 2.6 What each “sense” asks

Three different questions are routinely conflated under “which variables matter”. They are kept separate throughout.

**Sense V — variance-explaining.** Of the variation in a metric across cells, how much does each design factor account for, after the noise correction of (6)? This bounds every claim below it: a noise floor of 50% means no decomposition can explain more than half.

**Sense D — decision-relevant.** Does changing a factor change *which method you would choose*? Measured as the rate at which the best-performing method flips between adjacent factor levels, split into a raw rate and a gated rate that counts only flips whose margin survives a cross-fitted paired test at  $z = 2$ . The gap between them is the quantity of interest: it is how much of the apparent structure is noise.

**Sense G — diagnostic.** For a specific method, which factors hurt *it* relative to what was achievable? Measured by  $\kappa = (\text{method} - \text{baseline})/(\text{ceiling} - \text{baseline})$ , the fraction of available headroom captured, with SuSiE as baseline and PolyFun-oracle as ceiling.

Sense V and Sense D can disagree entirely, and in this dataset they do. A factor can move a metric a great deal while never changing the winner (every method degrades together), and a factor with a small main effect can flip the winner if it acts differently on different methods. Reporting only one of them overstates what is known.

## 3 Completeness: what is present and what is absent

Sixteen of the eighteen methods have all 1,125 cells populated with fit-failure rates between 0.0% and 0.5%. Two do not, and in both cases the absence is structural rather than a data defect:

- **Funmap** requires functional annotations and cannot run at all on the “none” arm (0/125 cells). It is complete (500/500) on both annotated arms. Its row on the unannotated stratum is *empty*, not *poor*, and no performance claim may be inferred from it.
- **FINEMAP-inf** is missing exactly 45 cells, all at  $\phi = 0.6$  and  $S = 1$  — nine rows  $\times$  five region sizes. It returns an all-NA posterior without raising an error when a single variant carries 60% of regional heritability. This is recorded through a dedicated `bad_pip` flag rather than being averaged away, so the loss is countable per method and per factor level.

CARMA was dropped from the method set mid-iteration on measured cost and is excluded from every analysis here; the eighteen methods reported are the complete set actually fitted.

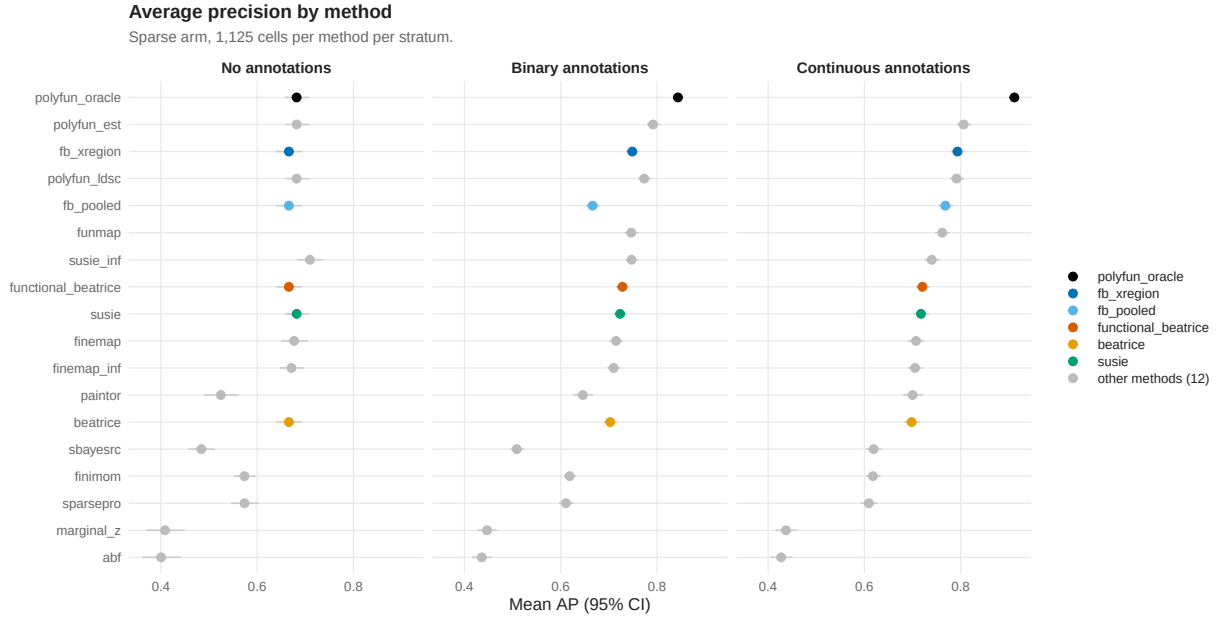


Figure 1: Mean average precision by method, with 95% intervals over the 1,125 cells of each stratum. PolyFun-oracle reads the simulated annotation weights and is a ceiling, not a deployable competitor.

## 4 Overall accuracy

Three features of Figure 1 are worth stating plainly.

**The oracle gap is real but not enormous.** PolyFun-oracle sits above every deployable method on the annotated strata, but the best estimated-prior methods recover a substantial share of that headroom. On the unannotated arm the oracle has no information to exploit and collapses onto SuSiE exactly — a mechanical check, not a coincidence (§11).

**Annotation-aware methods separate from annotation-blind ones only when annotations exist.** The ordering on the “none” panel is essentially a ranking of sparse-regression quality; the ordering on the annotated panels reorders substantially.

**The marginal baselines are informative.** `marginal_z` and `abf` sit far below everything else, which confirms the design poses a genuine fine-mapping problem: LD is doing real work, and simply ranking by marginal significance is not competitive.

## 5 When does each method do well?

The practical summary is that **sparsity is the binding constraint**. Moving from  $S = 1$  to  $S = 10$  costs far more accuracy than moving from  $\phi = 0.6$  to  $\phi = 0.05$ , and both dwarf region size and enrichment strength. For a practitioner this reframes the question: the useful prior information is how many independent signals a locus is likely to contain, not how large the locus is.

## 6 False discovery and calibration

The central trade-off in Figure 6 is that **ranking accuracy and calibration are not the same axis and the leaders differ**. Methods can be grouped:

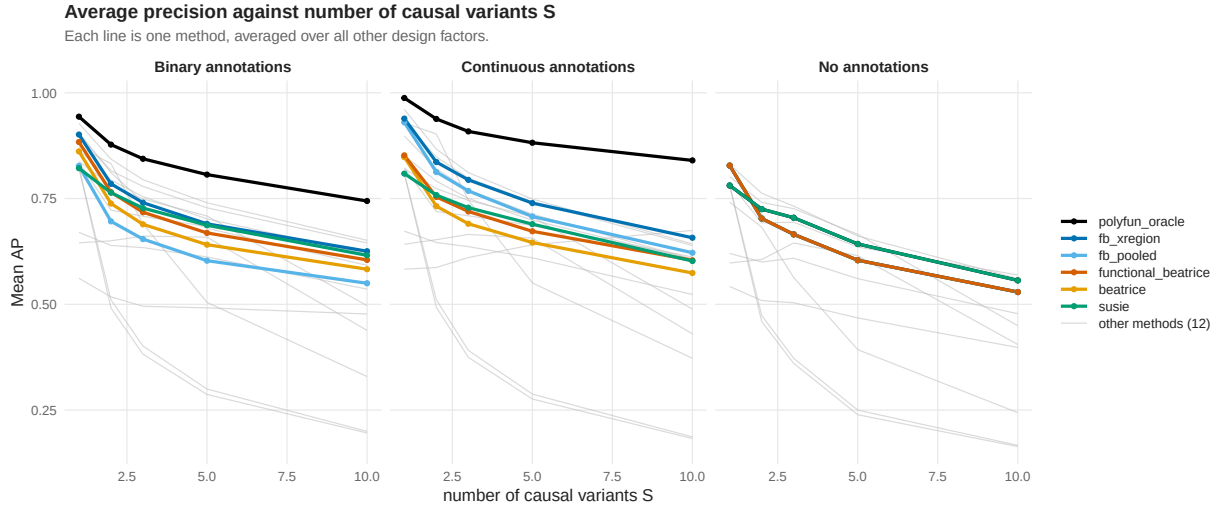


Figure 2: AP against the number of causal variants. Every method degrades as  $S$  grows; the spread between methods also widens.

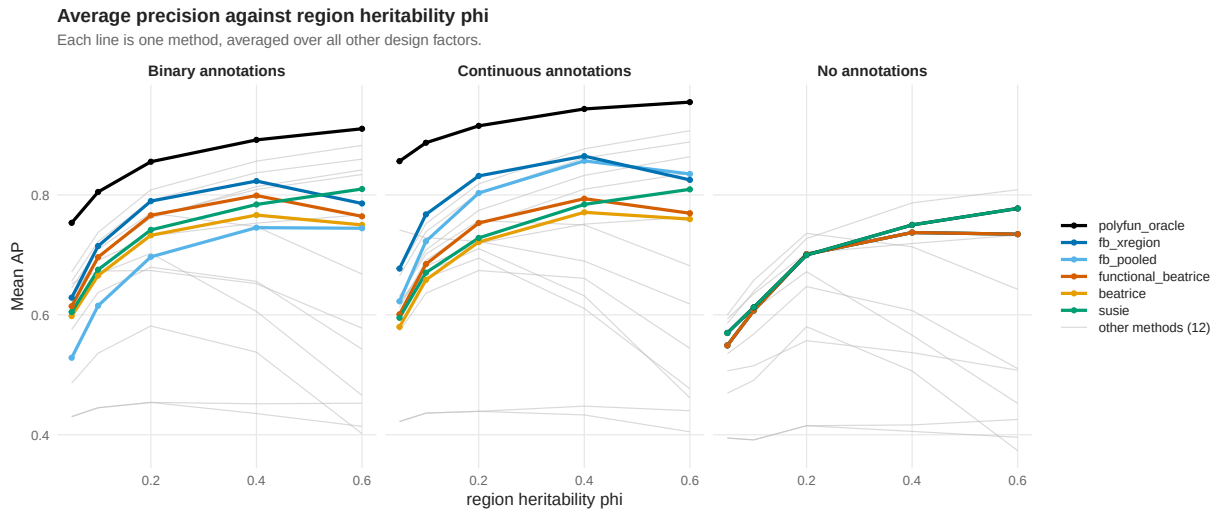


Figure 3: AP against region heritability. More signal helps everything, with diminishing returns at the top of the range.

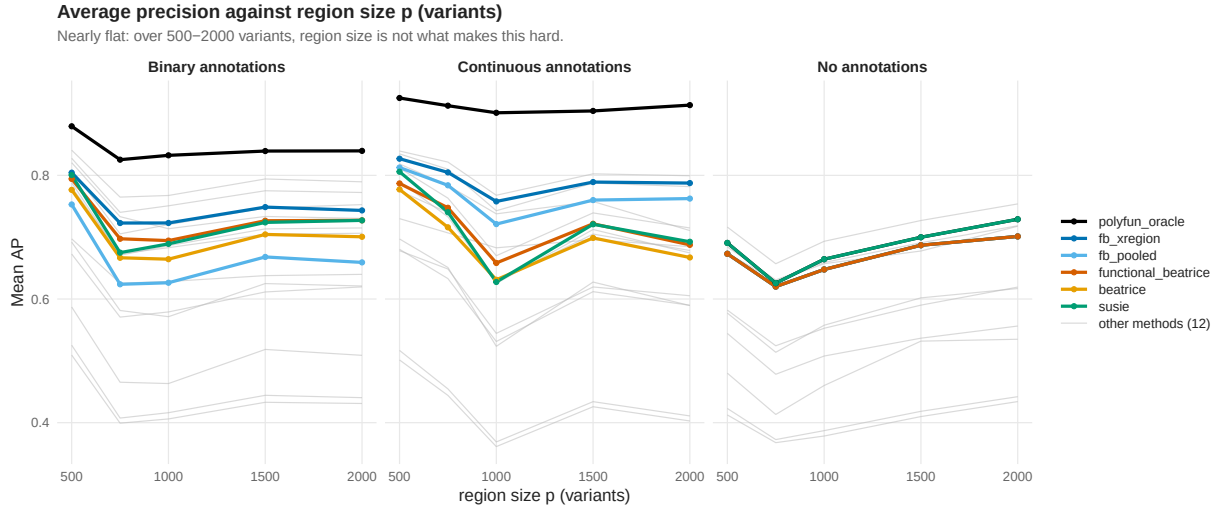


Figure 4: AP against region size. Nearly flat — over 500–2000 variants, region size is not what makes fine-mapping hard here.

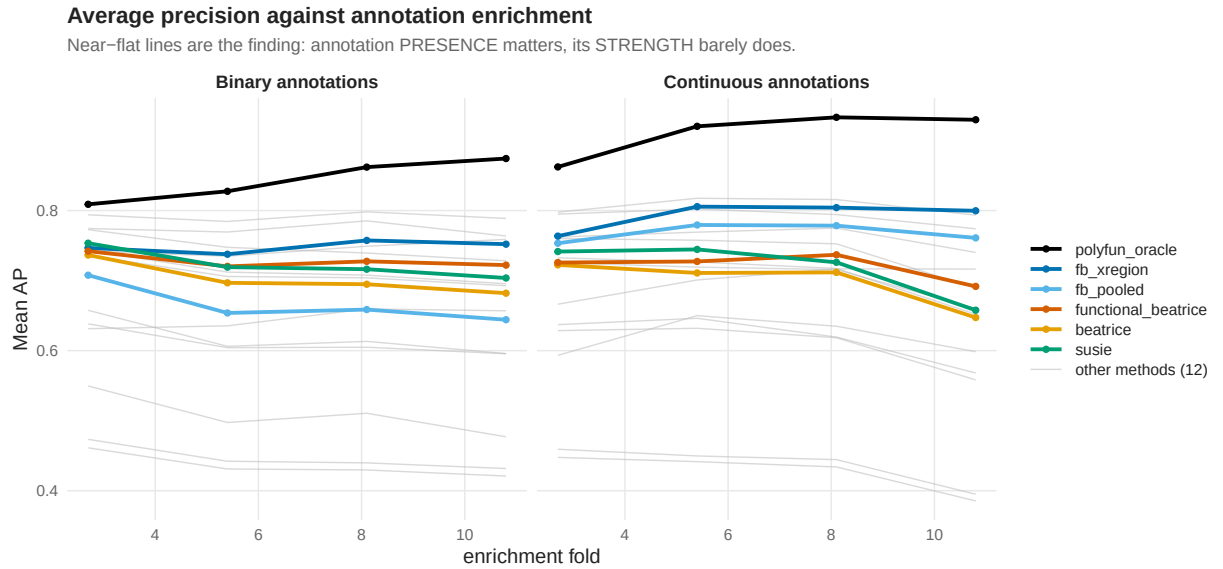


Figure 5: AP against annotation enrichment fold. The near-flat lines are the result: what matters is that annotations exist, not how strongly they are enriched over the range swept.



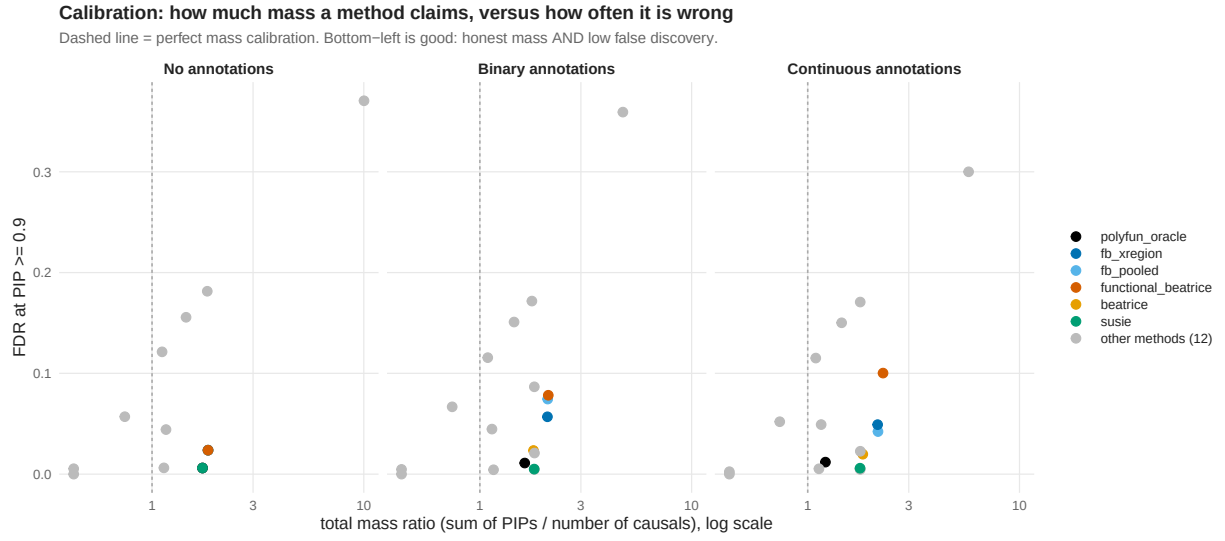


Figure 6: Mass claimed versus errors made. The dashed vertical line is perfect mass calibration; the bottom-left corner is the desirable region.

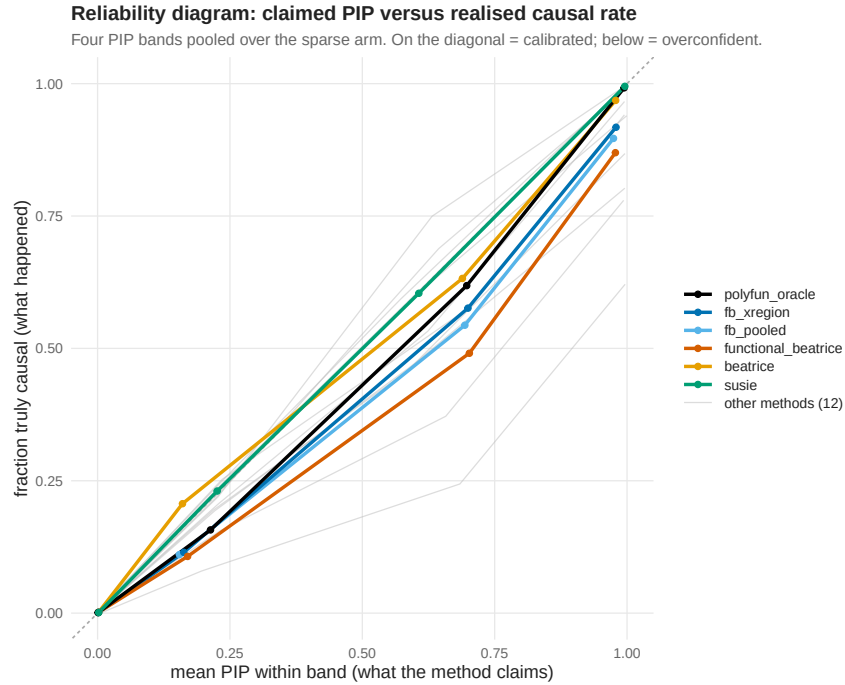


Figure 7: Reliability diagram over the whole sparse arm. Points below the diagonal are overconfident: the method claimed more probability than the realised causal rate justified.

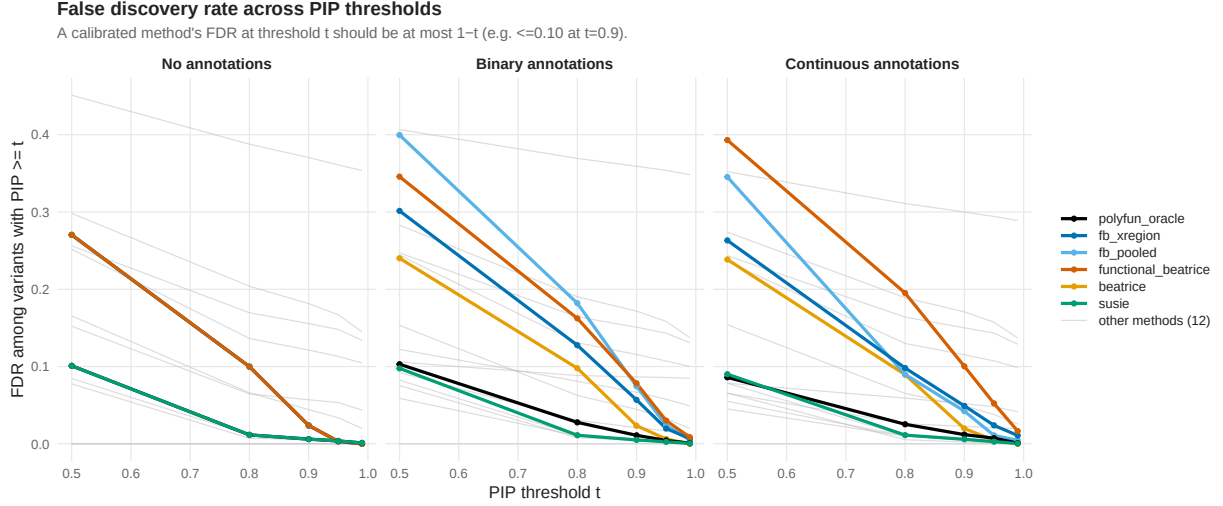


Figure 8: FDR as a function of the PIP threshold. A calibrated method should satisfy  $FDR_t \leq 1 - t$ .

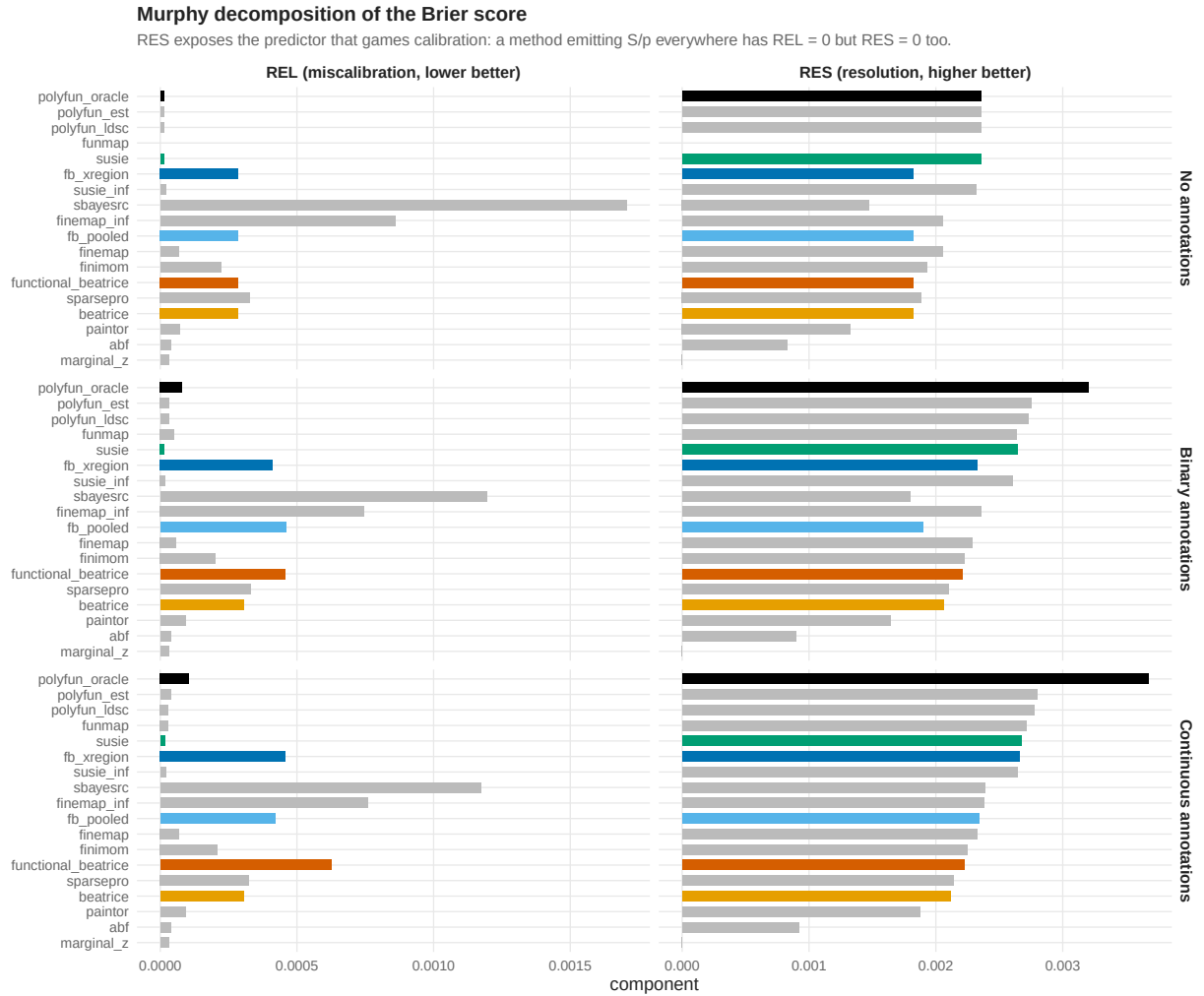


Figure 9: Murphy decomposition. REL is miscalibration (lower is better); RES is resolution (higher is better). RES is what separates a genuinely informative posterior from one that is calibrated by being vague.

- **Conservative and well-calibrated:** SuSiE-inf and FINEMAP hold mass ratios near 1.1–1.2 with low FDR, at some cost in AP.
- **Accurate and mass-inflated:** the BEATRICE family achieves high AP while claiming roughly twice the mass that exists, with FDR at  $\text{PIP} \geq 0.9$  an order of magnitude above the PolyFun family.
- **Diffuse:** sbayesrc spreads mass extremely widely (ratios in the multiples), which shows up as both poor calibration and a large credible-set burden.
- **Abstaining:** marginal\_z and abf place very little mass anywhere; their apparent FDR advantage is a consequence of rarely crossing the threshold at all, which is exactly why  $n_t$  is reported alongside.

## 7 Credible sets: the follow-up burden

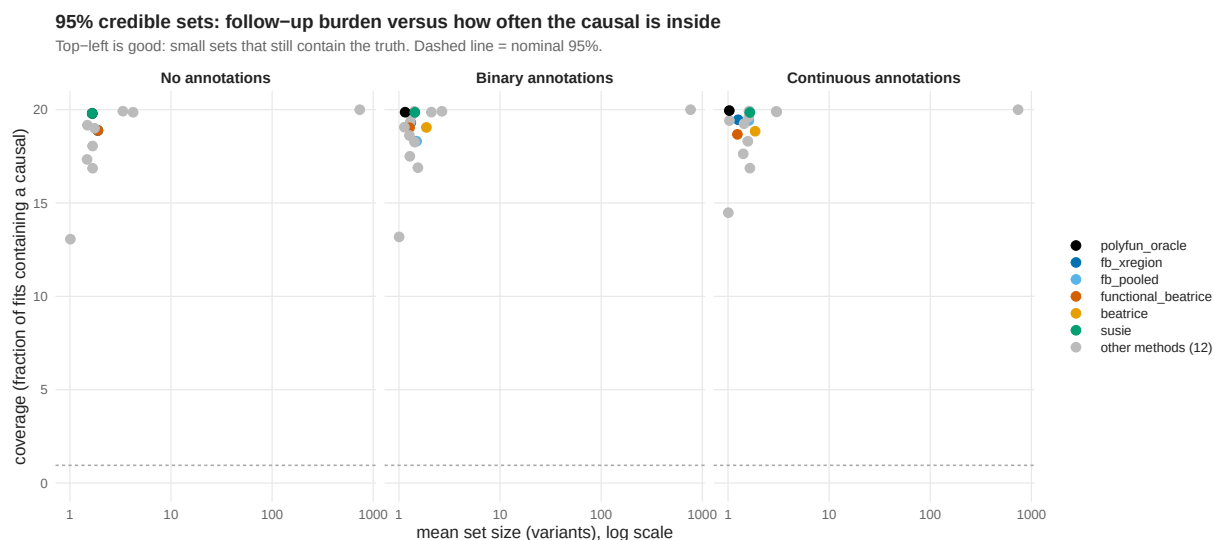


Figure 10: 95% credible sets. The practitioner’s question is the pair: how many variants must be taken to the bench, and how often is the causal one among them.

Coverage below the nominal 95% combined with small sets indicates a posterior that is too sharp; coverage at nominal with very large sets indicates one that is too diffuse. Neither number alone identifies which situation a method is in, which is why Figure 10 plots them jointly and on a log size axis.

## 8 Sense V: what moves the metric

Averaged over methods within stratum, the main-effect shares of AP are:  $S$  between 38.5% and 46.6%;  $\phi$  between 19.0% and 21.4%; region size 4.1–4.5%; enrichment fold 1.2–6.1%. The largest interaction is  $S:\phi$ , worth roughly 12–14%, which is interpretable: heritability per causal variant,  $\phi/S$ , is closer to the quantity that actually governs difficulty than either factor alone.

Because main effects carry the majority of the explained variance, **the marginal plots in §5 are honest summaries** — claims do not have to be stated conditionally. Had two-way terms dominated, they would.

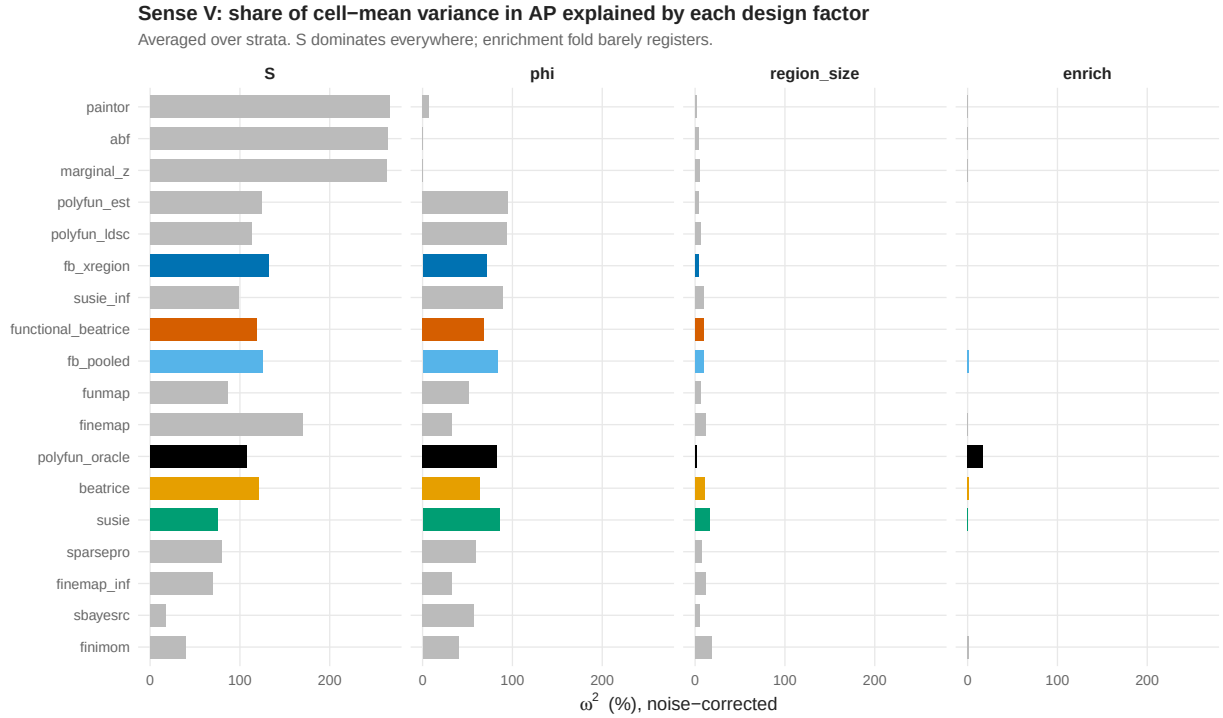


Figure 11: Noise-corrected  $\omega^2$  for each main effect, per method. *S* dominates; enrichment fold is negligible.

**A caveat that must travel with these numbers.**  $\omega^2$  is defined relative to the input distribution, here uniform over the levels swept. Widening a factor’s range mechanically inflates its importance. *S* spans a tenfold range (1–10) while enrichment spans fourfold (2.7–10.8), so part of *S*’s dominance is range. The correct phrasing is always “*S* explains 43% of the variance *over the range swept*”, never “*S* is the most important factor”.

## 9 Sense D: what changes the decision

This is the most cautionary result in the report. Raw flip rates are enormous — 64% to 100% of adjacent level-pairs show a change in which method scores best — while the gated rates collapse to near zero, and only 1.6% to 7.8% of cells have any statistically decidable winner at all.

**At 20 fits per cell, the benchmark cannot resolve which of 18 methods is best within a single cell.** Regime maps built from modal winners are therefore mostly empty, and the ones that are populated should not be read as recommendations.

This does *not* contradict §10. Those are different questions with very different power: “which of 18 methods wins this one cell” uses 20 fits, whereas “does method A beat method B on average” pools thousands of paired replicates. The benchmark answers the second question decisively and the first one barely at all — and reporting both is the honest position.

## 10 Sense G: headroom and the Functional BEATRICE ablation

Because all methods run on the same simulated data, differences can be taken *within* a replicate, which removes the between-scenario variance that dominates the marginal comparisons and is what gives this analysis its power.

Table 4 reports the paired contrasts. Both stages of the ablation are significant, and the

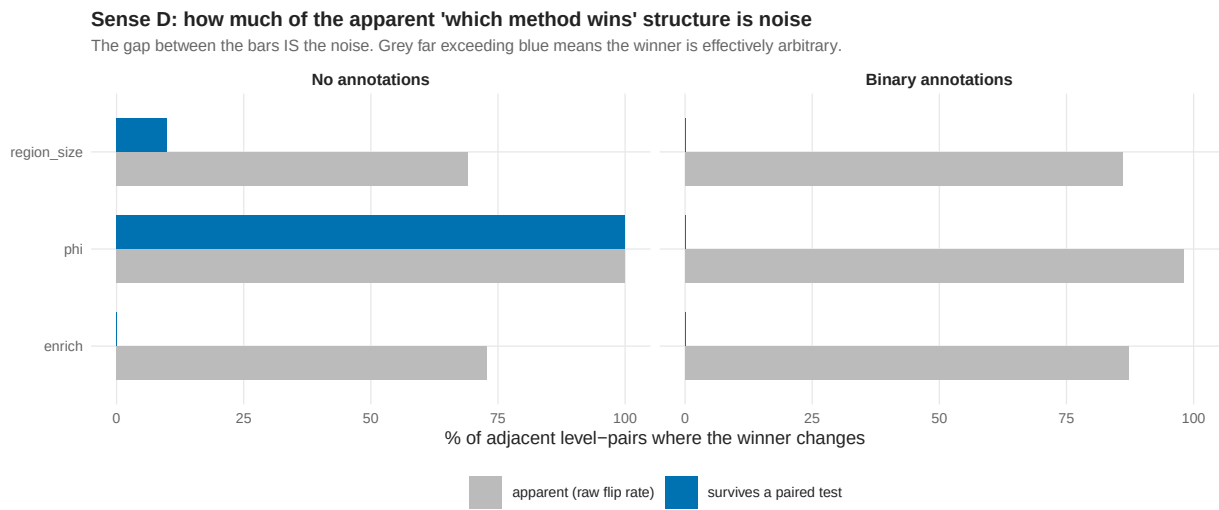


Figure 12: Apparent versus statistically supported changes of winner. The gap is the noise.

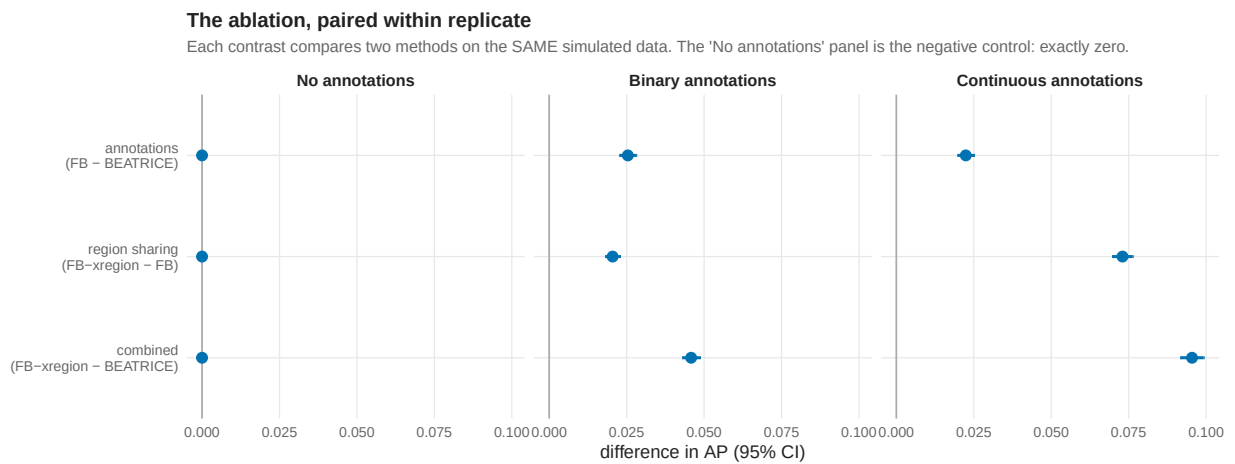


Figure 13: The ablation, paired within replicate. Each contrast compares two methods on identical simulated data. The “No annotations” panel is the negative control.

negative control is exact:

- **Annotations** (Functional BEATRICE – BEATRICE) add +0.025 AP under binary annotations ( $t = 18.4$ ) and +0.022 under continuous ( $t = 16.4$ ).
- **Region sharing** (FB-xregion – Functional BEATRICE) adds a further +0.021 ( $t = 16.8$ ) and +0.073 ( $t = 43.5$ ) respectively.
- **Combined**, +0.046 and +0.095 AP.
- On the unannotated arm all three contrasts are **exactly**  $0.0000 \pm 0.0000$  over 1,250 paired replicates.

That last point deserves emphasis. With no annotations to exploit, the three variants are bit-identical, so the measured gains cannot be an artefact of the extra machinery, of additional parameters, or of run-to-run variation. It is as clean a negative control as the design admits.

Notably, **region sharing contributes more than annotations alone under continuous annotations** — 0.073 against 0.022 — so the extension is not riding on the base method.

**$\kappa$  is reported but should not be over-read.** The headroom denominator (ceiling – baseline) is small in many cells, and after gating out cells with headroom  $\leq 0.02$  the residual  $\kappa$  estimates carry standard deviations of 1.17–1.29 against means between  $-0.11$  and  $0.31$ . On the unannotated arm the gate removes everything, correctly: with no annotations the ceiling and the baseline are the same method and  $\kappa$  is undefined by construction. We report  $\kappa$  for completeness and rest no conclusion on it.

## 11 Validity checks

The following are verified mechanically on every run and all pass for the sparse arm:

1. In-sample LD only — zero rows with a reference-panel size.
2. Nine job rows present, matching the grid.
3. Balanced cells per method after documented exclusions — 1,125 for every analysed method.
4. No method entirely absent outside the “none” arm.
5. `polyfun.est = susie` on the unannotated arm.
6. `polyfun.ldsc = susie` on the unannotated arm.

Checks 5 and 6 are the strongest structural tests available: both methods reduce to SuSiE analytically when there are no annotations, so any discrepancy would indicate a fault in the annotation plumbing rather than in a method. They agree to numerical precision.

## 12 Limitations

1. **In-sample LD only.** Iteration 003 established that in-sample and reference-panel LD can *invert* method rankings. Nothing here transfers to the reference-panel setting.
2. **Sparse architecture only.** This report covers the arm with no infinitesimal background. The sparse-plus-infinitesimal arm is a separate analysis.

3. **Range dependence.** See §8; all importance statements are relative to the ranges swept.
4. **Per-cell decisions are underpowered.** See §9.
5.  **$\kappa$  decomposition unavailable.** The replicate-level table does not carry the region index, so  $\sigma_u^2$  is unidentifiable for the derived  $\kappa$  and  $\Delta$  responses. The corrected decomposition is skipped rather than reported uncorrected, which would inflate whole-plot terms by  $m\sigma_u^2/2$  per degree of freedom.
6. **One tuning configuration.** BEATRICE-family hyperparameters come from a single Optuna study and are held fixed; competitors run at their documented defaults.

### 13 Full results tables

Bold rows are the BEATRICE family. *cells* counts non-missing cells out of 1,125.

Table 1: **No annotations** — all methods, sparse arm. Ranked by mean AP. *cells* is the number of non-missing cells out of 1,125; anything below that is a genuine absence, explained in §3.

| method                     | AP     | FDR@.9 | mass  | ECE <sub>hi</sub> | CS <sub>95</sub> | cov <sub>95</sub> | RES    | BSS    | cells |
|----------------------------|--------|--------|-------|-------------------|------------------|-------------------|--------|--------|-------|
| susie_inf                  | 0.7095 | 0.0062 | 1.14  | 0.0288            | 4.2              | 19.864            | 0.0023 | 0.575  | 125   |
| polyfun_est                | 0.6819 | 0.0061 | 1.73  | 0.0225            | 1.7              | 19.800            | 0.0024 | 0.579  | 125   |
| polyfun_ldsc               | 0.6819 | 0.0061 | 1.73  | 0.0225            | 1.7              | 19.800            | 0.0024 | 0.579  | 125   |
| polyfun_oracle             | 0.6819 | 0.0061 | 1.73  | 0.0225            | 1.7              | 19.800            | 0.0024 | 0.579  | 125   |
| susie                      | 0.6819 | 0.0061 | 1.73  | 0.0225            | 1.7              | 19.800            | 0.0024 | 0.579  | 125   |
| finemap                    | 0.6767 | 0.0442 | 1.16  | 0.0683            | 1.5              | 19.168            | 0.0021 | 0.490  | 125   |
| finemap_inf                | 0.6711 | 0.1815 | 1.82  | 0.1619            | 1.7              | 16.864            | 0.0021 | 0.056  | 120   |
| <b>beatrice</b>            | 0.6659 | 0.0237 | 1.84  | 0.1434            | 1.9              | 18.888            | 0.0018 | 0.321  | 125   |
| <b>fb_pooled</b>           | 0.6659 | 0.0237 | 1.84  | 0.1434            | 1.9              | 18.888            | 0.0018 | 0.321  | 125   |
| <b>fb_xregion</b>          | 0.6659 | 0.0237 | 1.84  | 0.1434            | 1.9              | 18.888            | 0.0018 | 0.321  | 125   |
| <b>functional_beatrice</b> | 0.6659 | 0.0237 | 1.84  | 0.1434            | 1.9              | 18.888            | 0.0018 | 0.321  | 125   |
| sparsepro                  | 0.5737 | 0.1556 | 1.45  | 0.1232            | 1.5              | 17.336            | 0.0019 | 0.249  | 125   |
| finimom                    | 0.5735 | 0.1213 | 1.12  | 0.0890            | 1.7              | 18.056            | 0.0019 | 0.322  | 125   |
| paintor                    | 0.5247 | 0.0570 | 0.74  | 0.0615            | 1.8              | 19.008            | 0.0013 | 0.372  | 125   |
| sbayesrc                   | 0.4841 | 0.3705 | 10.02 | 0.2207            | 1.0              | 13.064            | 0.0015 | -0.552 | 125   |
| marginal_z                 | 0.4086 | 0.0000 | 0.43  | NaN               | 735.6            | 20.000            | 0.0000 | -0.003 | 125   |
| abf                        | 0.4005 | 0.0054 | 0.43  | 0.0237            | 3.3              | 19.920            | 0.0008 | 0.314  | 125   |
| funmap                     | NaN    | 0.0000 | NaN   | NaN               | NaN              | 0.000             | NaN    | NaN    | 0     |

Table 2: **Binary annotations** — all methods, sparse arm. Ranked by mean AP. *cells* is the number of non-missing cells out of 1,125; anything below that is a genuine absence, explained in §3.

| method                     | AP     | FDR@.9 | mass | $ECE_{hi}$ | $ CS_{95} $ | cov <sub>95</sub> | RES    | BSS    | cells |
|----------------------------|--------|--------|------|------------|-------------|-------------------|--------|--------|-------|
| polyfun_oracle             | 0.8432 | 0.0110 | 1.63 | 0.0715     | 1.2         | 19.866            | 0.0032 | 0.694  | 500   |
| polyfun_est                | 0.7914 | 0.0046 | 1.81 | 0.0376     | 1.4         | 19.914            | 0.0028 | 0.644  | 500   |
| polyfun_ldsc               | 0.7732 | 0.0209 | 1.81 | 0.0440     | 1.3         | 19.668            | 0.0027 | 0.635  | 500   |
| <b>fb_xregion</b>          | 0.7484 | 0.0569 | 2.08 | 0.1454     | 1.3         | 19.306            | 0.0023 | 0.301  | 500   |
| susie_inf                  | 0.7470 | 0.0043 | 1.16 | 0.0283     | 2.1         | 19.870            | 0.0026 | 0.620  | 500   |
| funmap                     | 0.7462 | 0.0866 | 1.81 | 0.0746     | 1.3         | 18.616            | 0.0026 | 0.614  | 500   |
| <b>functional_beatrice</b> | 0.7280 | 0.0783 | 2.10 | 0.1547     | 1.3         | 19.034            | 0.0022 | 0.245  | 500   |
| susie                      | 0.7232 | 0.0050 | 1.80 | 0.0218     | 1.4         | 19.856            | 0.0026 | 0.624  | 500   |
| finemap                    | 0.7147 | 0.0447 | 1.14 | 0.0648     | 1.3         | 19.346            | 0.0023 | 0.535  | 500   |
| finemap_inf                | 0.7096 | 0.1717 | 1.76 | 0.1585     | 1.5         | 16.894            | 0.0024 | 0.141  | 480   |
| <b>beatrice</b>            | 0.7026 | 0.0235 | 1.79 | 0.1592     | 1.9         | 19.050            | 0.0021 | 0.369  | 500   |
| <b>fb_pooled</b>           | 0.6661 | 0.0745 | 2.09 | 0.1547     | 1.5         | 18.312            | 0.0019 | 0.187  | 500   |
| paintor                    | 0.6458 | 0.0668 | 0.74 | 0.0919     | 1.1         | 19.064            | 0.0016 | 0.452  | 500   |
| finimom                    | 0.6182 | 0.1156 | 1.09 | 0.0876     | 1.4         | 18.264            | 0.0022 | 0.378  | 500   |
| sparsepro                  | 0.6107 | 0.1509 | 1.45 | 0.1232     | 1.3         | 17.504            | 0.0021 | 0.286  | 500   |
| sbayesrc                   | 0.5086 | 0.3592 | 4.74 | 0.2470     | 1.0         | 13.186            | 0.0018 | -0.313 | 500   |
| marginal_z                 | 0.4467 | 0.0000 | 0.43 | NaN        | 765.5       | 20.000            | 0.0000 | -0.003 | 500   |
| abf                        | 0.4357 | 0.0047 | 0.43 | 0.0255     | 2.7         | 19.914            | 0.0009 | 0.334  | 500   |

Table 3: **Continuous annotations** — all methods, sparse arm. Ranked by mean AP. *cells* is the number of non-missing cells out of 1,125; anything below that is a genuine absence, explained in §3.

| method                     | AP     | FDR@.9 | mass | $ECE_{hi}$ | $ CS_{95} $ | cov <sub>95</sub> | RES    | BSS    | cells |
|----------------------------|--------|--------|------|------------|-------------|-------------------|--------|--------|-------|
| polyfun_oracle             | 0.9114 | 0.0119 | 1.21 | 0.0791     | 1.0         | 19.952            | 0.0037 | 0.774  | 500   |
| polyfun_est                | 0.8061 | 0.0051 | 1.77 | 0.0414     | 1.6         | 19.928            | 0.0028 | 0.641  | 500   |
| <b>fb_xregion</b>          | 0.7933 | 0.0491 | 2.14 | 0.1529     | 1.3         | 19.462            | 0.0027 | 0.327  | 500   |
| polyfun_ldsc               | 0.7914 | 0.0047 | 1.77 | 0.0360     | 1.6         | 19.906            | 0.0028 | 0.638  | 500   |
| <b>fb_pooled</b>           | 0.7681 | 0.0422 | 2.15 | 0.1504     | 1.6         | 19.424            | 0.0023 | 0.278  | 500   |
| funmap                     | 0.7617 | 0.0227 | 1.77 | 0.0367     | 1.6         | 19.610            | 0.0027 | 0.628  | 500   |
| susie_inf                  | 0.7398 | 0.0053 | 1.13 | 0.0284     | 3.0         | 19.886            | 0.0026 | 0.618  | 500   |
| <b>functional_beatrice</b> | 0.7204 | 0.1002 | 2.27 | 0.1728     | 1.2         | 18.680            | 0.0022 | 0.117  | 500   |
| susie                      | 0.7175 | 0.0059 | 1.77 | 0.0236     | 1.6         | 19.852            | 0.0027 | 0.621  | 500   |
| finemap                    | 0.7071 | 0.0492 | 1.16 | 0.0682     | 1.4         | 19.250            | 0.0023 | 0.529  | 500   |
| finemap_inf                | 0.7053 | 0.1708 | 1.77 | 0.1555     | 1.6         | 16.870            | 0.0024 | 0.143  | 480   |
| paintor                    | 0.7001 | 0.0520 | 0.74 | 0.0790     | 1.0         | 19.420            | 0.0019 | 0.520  | 500   |
| <b>beatrice</b>            | 0.6981 | 0.0198 | 1.82 | 0.1531     | 1.9         | 18.846            | 0.0021 | 0.362  | 500   |
| sbayesrc                   | 0.6192 | 0.3000 | 5.75 | 0.1993     | 1.0         | 14.482            | 0.0024 | -0.254 | 500   |
| finimom                    | 0.6177 | 0.1152 | 1.09 | 0.0910     | 1.6         | 18.308            | 0.0022 | 0.378  | 500   |
| sparsepro                  | 0.6093 | 0.1502 | 1.45 | 0.1236     | 1.4         | 17.636            | 0.0021 | 0.294  | 500   |
| marginal_z                 | 0.4371 | 0.0000 | 0.43 | NaN        | 738.4       | 20.000            | 0.0000 | -0.003 | 500   |
| abf                        | 0.4271 | 0.0024 | 0.43 | 0.0237     | 3.0         | 19.910            | 0.0009 | 0.333  | 500   |



Table 4: Paired ablation on AP. Each row compares two methods on the *same* simulated replicate;  $n$  is the number of paired replicates.  $t = \bar{d}/\text{SE}(\bar{d})$ .

| stratum                | contrast       | $\bar{d}$ | sd     | $n$  | SE      | $t$  |
|------------------------|----------------|-----------|--------|------|---------|------|
| No annotations         | annotations    | 0.0000    | 0.0000 | 1250 | 0.00000 | —    |
| No annotations         | region sharing | 0.0000    | 0.0000 | 1250 | 0.00000 | —    |
| No annotations         | combined       | 0.0000    | 0.0000 | 1250 | 0.00000 | —    |
| Binary annotations     | annotations    | 0.0254    | 0.0975 | 4981 | 0.00138 | 18.4 |
| Binary annotations     | region sharing | 0.0205    | 0.0859 | 4980 | 0.00122 | 16.8 |
| Binary annotations     | combined       | 0.0458    | 0.1026 | 4980 | 0.00145 | 31.5 |
| Continuous annotations | annotations    | 0.0224    | 0.0962 | 4965 | 0.00136 | 16.4 |
| Continuous annotations | region sharing | 0.0730    | 0.1183 | 4960 | 0.00168 | 43.5 |
| Continuous annotations | combined       | 0.0954    | 0.1352 | 4960 | 0.00192 | 49.7 |